



You said it...



Thanks for the article on libraries. Some of them are really helpful—it helps me to understand the importance of some coding -- for example, extern 'C'.

—Vineesh Kumar, by e-mail to Nileshe Govande, on his article on Libraries published on Page 66 in the December 2008 issue



First of all, thanks for this wonderful article—it really helped me a lot. I am new to the C programming language and my question may seem pretty naive but some help would be really great. When you described the process of writing *main.c* for a dynamic library, you showed it with different APIs.

My question is: When we call *printf()* in the normal way in, say, *example.c*, even then it is dynamically linked. Am I right? But in that case we just call it *printf()*, with no use of APIs. Can you please spare a few moments of your time to explain the difference, or suggest some reading material so I can equip myself with sufficient knowledge before proceeding?

—Himanshu Mall, by e-mail to Nileshe Govande on his article on Libraries

Nileshe replies: To answer your question on whether when *printf()* is called the normal way, is it even then dynamically linked...you are absolutely right, it is. In fact, *printf()* being part of *libc* will always get linked dynamically unless you specify *-static* at compile time. When you wrote, "But then there we just call it

printf(); no use of APIs," did you mean to say APIs like *dlopen()* and *dlsym()*?

We need to do these acrobatics because we are linking a non-standard C library and the symbol address of the 'display' function is not known to the compiler. Since *glibc* is present with the compiler itself, the address resolution is not required. Going further, if you were able to build the 'display' executable in the dynamic library section, try:

```
[root@localhost dynamic]# nm -u display
U _dlclose@@GLIBC_2.0
U _dlerror@@GLIBC_2.0
U _dlopen@@GLIBC_2.1
U _dlsym@@GLIBC_2.0
U _exit@@GLIBC_2.0
U _fprintf@@GLIBC_2.0
w __gmon_start__
w _jv_RegisterClasses
U __libc_start_main@@GLIBC_2.0
```

So the functions *dlopen()*, *dlsym()*, etc, are present in your *libc* itself. Even:

```
/******a.c*****/
#include <stdio.h>
int main()
{
    printf("Hello!!!\n");
    return 0;
}
#gcc a.c
#./a.out
Hello!!!

#gcc -v a.c
```

Hence, even the linking of *glibc* is invisible to us. But if you really want to view it, try:

Now, notice the output!



I'm studying in the 10th standard and want to enhance my knowledge of Linux. Thanks for making me a Linux geek. I've been a

regular reader of the magazine since August 2008.

I'm currently using Ubuntu Ultimate and I've a request: can you include Ubuntu Ultimate 2009 and Mandriva Powerpack 2009 in the LFY DVD? As an Ubuntu fan, I'm very passionate about Ubuntu Ultimate and also Mandriva. Since I don't have broadband connectivity, I can't download these images.

—Sarith Mohan, by e-mail

ED: It's great to know that LFY is helping you in your journey with Linux :-). Mandriva 'Free' 2009 was bundled with the November issue. Check it out! The Mandriva Powerpack editions are not freely distributable. As for the Ubuntu Ultimate edition, it was released after our Ubuntu multi-boot DVD was packed. So, we couldn't bundle it. Let's hope they release the Ultimate edition on time for v9.04. We'll surely try to bundle it then.

Errata

Misprints in December 2008 issue:

- Pg 37: In column 1, first paragraph, the spelling of Kasargod was spelled as Kazargode.
- Pg 37: Anoop John's name was misspelled as Anoop Johnson
- Pg 37: It was a 44-day long Freedom Walk, and not 43 days long as printed.
- Pg 88: In column 2, the second command snippet reads *create table session; use session; It should have read create database session; use session;*
- Pg 89: In column 2, source code for login.php reads:

```
$con=mysql_connect('127.0.0.1','root','slavasi
va') or die(mysql_error());
mysql_select_db('session',$con)
or die(mysql_error());
It should have read:
$con=mysql_connect('127.0.0.1','user','pass')
or die(mysql_error());
mysql_select_db('session',$con)
or die(mysql_error());
```

Please send your comments or suggestions to:

The Editor

LINUX FOR YOU Magazine

D-87/1, Okhla Industrial Area,

Phase I, New Delhi 110020

Ph: 011-26810601/02/03, Fax: 26817563

e-mail: lfyouedit@yindia.com

Website: www.openTis.com